

Analysis of Neural Correlates and Content Personalization in Neuromarketing

1. Introduction and Objectives

The primary objective of this study is to bridge the gap between physiological signals and consumer decision-making processes. By analyzing the DEAP dataset, which contains electroencephalogram (EEG) and peripheral physiological recordings, we aim to construct a predictive framework that correlates biological responses with emotional states—specifically Valence and Arousal—and subsequent consumer behavior, modeled here as "Liking" or purchase intent. The study further seeks to evaluate the efficacy of personalized marketing content by identifying specific cognitive states that maximize engagement for different categories of advertising material.

2. Study of Correlations: Signal-Emotion-Behavior Dynamics

2.1 Physiological Indicators of Emotion

The core of our analysis relies on extracting spectral power densities from 32 EEG channels and 8 peripheral channels. The data processing pipeline isolates specific frequency bands—Theta (4-8 Hz), Alpha (8-12 Hz), Beta (12-25 Hz), and Gamma (25-45 Hz)—which are well-documented correlates of cognitive load and emotional processing.

Peripheral signals provide critical context for these neural oscillations. For instance, Galvanic Skin Response (GSR) serves as a robust indicator of emotional intensity or Arousal, tracking sweat gland activity linked to the sympathetic nervous system. Similarly, electromyography (EMG) channels, such as the Zygomaticus Major (zEMG) and Trapezius (tEMG), capture somatic manifestations of emotion, identifying positive valence through subtle smiling mechanisms and stress through neck tension, respectively.

2.2 Mapping Signals to Consumer Behavior

To model consumer behavior, the "Liking" label (rated on a 1–9 scale) serves as the primary proxy for purchase intent. We established a binary classification threshold where a Liking score greater than 5.0 indicates a positive "Buy" decision.

Using a Random Forest Classifier trained on these physiological features, the model demonstrated a significant ability to predict this purchase intent. For the test subject (Subject 01), the model achieved an accuracy of 98.13%, successfully distinguishing between positive and negative consumer intent based solely on biological signals. This high degree of accuracy suggests that consumer preference is not merely a psychological

abstraction but is rooted in detectable, consistent physiological patterns that precede conscious reporting. The strong correlation implies that distinct ranges of Valence (positivity) and Arousal (excitement) act as precursors to the "Liking" response.

3. Comparative Evaluation of Personalized Content

3.1 Categorization of Marketing Stimuli

To evaluate the impact of personalization, marketing content was segmented into four distinct emotional archetypes based on median splits of Valence and Arousal ratings. This categorization allows for a comparative analysis of how different cognitive states influence the reception of specific ad types:

- **Action/High-Octane:** Characterized by above-average Valence and Arousal.
- **Sentimental/Warm:** Defined by above-average Valence but below-average Arousal.
- **Fear/Problem-Solution:** Defined by below-average Valence but high Arousal.
- **Melancholic/Serious:** Characterized by low levels of both Valence and Arousal.

3.2 Effectiveness of State-Dependent Personalization

The analysis reveals that "Liking"—and by extension, purchase intent—is not maximized by simply increasing excitement (Arousal) or positivity (Valence) indiscriminately. Instead, effectiveness is contingent upon aligning the user's current cognitive state with the specific "sweet spot" of the content category.

For example, our analysis of top-performing content indicates that **Sentimental/Warm** advertisements achieve maximum efficacy when the user's Valence is moderately high (between 3.59 and 4.23) and Arousal is contained within a narrower, lower range (3.12 to 3.31). Conversely, **Fear/Problem-Solution** narratives require a distinct physiological state to drive conversion, thriving when the user experiences lower Valence (2.89–3.16) combined with significantly higher Arousal (4.20–5.18).

This demonstrates that personalized content, which adapts to the user's real-time physiological baseline, is theoretically superior to static content delivery. By targeting these specific "optimal ranges," marketers can ensure that a "High-Octane" message is delivered only when the consumer is physiologically primed for excitement, thereby maximizing the probability of a "Like" response.

4. Key Findings and Strategic Recommendations

4.1 Synthesis of Results

The study confirms that consumer behavior can be accurately predicted from physiological signals with high precision using non-linear classifiers like Random Forest. Furthermore, we identified that the effectiveness of marketing stimuli is highly sensitive to the viewer's pre-existing emotional state. There is no universal "buying state"; rather, there are unique physiological windows for every genre of content that maximize the likelihood of a positive purchase decision.

4.2 Recommendations for Implementation

To operationalize these findings, we propose a dynamic system that moves beyond static classification. The next phase of development should focus on training a regression model to predict continuous Valence and Arousal values in real-time.

This system should calculate a "distance score" or error metric that quantifies how far a user's current state deviates from the identified optimal range for a given ad campaign. This metric can serve as a gating mechanism, ensuring ads are only displayed when the distance to the optimal state approaches zero. Finally, to scale this personalization, we recommend converting these bio-signals into tokenized inputs for a Large Language Model (LLM). By feeding the LLM with real-time "distance scores" and Liking probabilities, the system can dynamically generate or select content that bridges the gap between the user's current state and the optimal state required for conversion.